AYE
TECH HUB

• HVAC ENGINEERING • LESSON 08

Air Distribution Systems

Diffusers · Grilles · VAV Boxes · FCUs · ACH

Lesson 08 of 30 | HVAC Engineering Program

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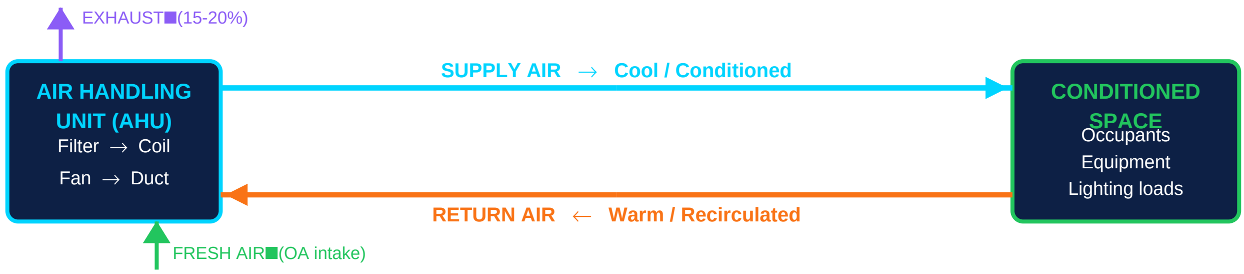
- Supply, return, and exhaust air — the complete air circulation loop
- Diffusers, grilles, and registers — types, selection, and placement rules
- CAV vs VAV: constant volume vs variable volume air systems explained
- Throw, drop, and spread — how to read diffuser performance data
- Air Changes per Hour (ACH): what different spaces actually need

Air Distribution — Moving Conditioned Air Where It Is Needed

Air distribution is the final step of the HVAC system — delivering conditioned air (cooled or heated, filtered, humidified) from the Air Handling Unit (AHU) to every occupied zone in the building, then returning that air back for reconditioning. **Good air distribution = occupant comfort. Poor distribution = hot spots, cold drafts, noise complaints, and wasted energy.**

Figure 1 — The Air Distribution Loop: Supply · Return · Exhaust · Fresh Air



Return air makes up ~80-85% of supply air. The remaining 15-20% is exhausted and replaced by fresh outdoor air.

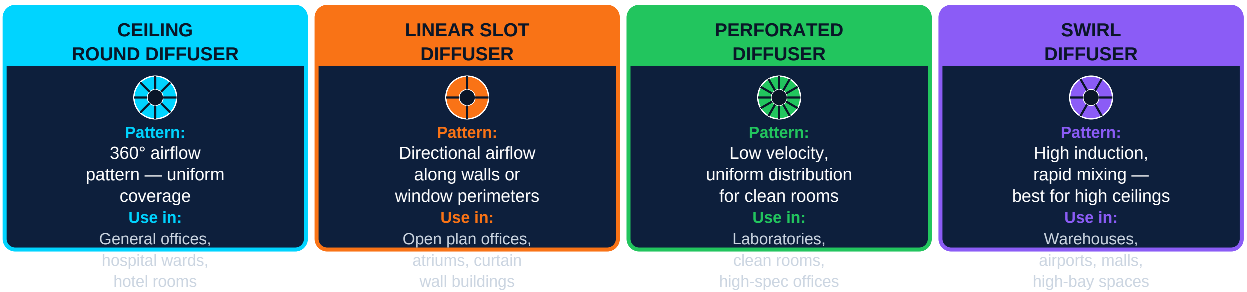
Air Stream	Direction	Typical % of Supply	Key Point
Supply Air	AHU → Ductwork → Diffusers → Room	100% (the reference)	Conditioned: filtered, cooled or heated, dehumidified
Return Air	Room → Return Grilles → AHU	80–85% of supply	Recirculated for reconditioning — reduces energy waste
Exhaust Air	Room → Exhaust Grilles → Outside	15–20% of supply	Removes odours, CO■, and stale air from bathrooms, kitchens
Fresh / Outdoor Air	Outside → AHU mixing box → AHU	15–20% of supply (makeup)	ASHRAE 62.1 minimum ventilation: 5–10 L/s per person
Transfer Air	Corridor → Adjacent room (through gap under door)	Variable	Maintains positive pressure in corridors for infection control

Pressure cascade principle: In healthcare buildings, positive pressure rooms (operating theatres, isolation rooms for immune-compromised) push air outward to keep contaminants out. Negative pressure rooms (infectious disease isolation) pull air inward to contain pathogens. This is achieved by supplying more air than is exhausted (positive) or vice versa (negative).

Air Terminal Units — Diffusers, Grilles, and VAV Boxes

Air terminal units are the end points of the duct system — the devices that deliver air into the space or collect it for return. Selecting the wrong type for a space causes noise, drafts, and poor temperature control.

Figure 2 — Four Main Air Diffuser Types: Pattern, Application, and Selection Guide



Terminal Device	Function	Key Selection Criteria
Supply Diffuser (ceiling)	Delivers supply air into the room with controlled throw and induction. Creates a blanket of conditioned air at ceiling level.	Neck velocity (2-4 m/s max), throw distance, noise rating (NC), ceiling height, and airflow volume (L/s)
Return Air Grille	Collects room air to send back to AHU. No moving parts — passive opening with fixed or adjustable blades.	Face velocity must stay ≤ 1.5 m/s to avoid noise. Size for low pressure drop. Place away from supply diffusers.
Linear Slot Diffuser	Delivers air along a line — typically at perimeter windows or ceiling slots. Creates a curtain of air to counteract solar heat gain or cold from glazing.	Ideal at window perimeters in buildings with floor-to-ceiling glass. Handles high throw at low noise. Common in airports and offices.
VAV Box (Variable Air Volume)	Modulates the airflow to a zone based on thermostat demand. Contains a damper, actuator, and often a reheat coil.	Size for peak cooling load but verify minimum airflow for ventilation. Reheat VAV for heating-dominated zones. Fan-powered VAV for high-ceiling spaces.
Fan Coil Unit (FCU)	Self-contained unit with its own fan and coil — provides local conditioning without connecting to a central AHU duct system.	Used in hotel rooms, apartments, perimeter zones. Two-pipe (cooling only) or four-pipe (cooling + heating) configurations.
Exhaust / Transfer Grille	Removes air from toilets, kitchens, server rooms, or passes air from one room to another through a wall opening.	Size for face velocity < 2 m/s. Use acoustic lining for grilles between occupied spaces. Ensure fire damper compliance.

Throw, Drop, and Spread — Reading Diffuser Performance Data

Term	Definition	Typical Design Target
Throw	Horizontal distance the air jet travels from the diffuser to the point where velocity drops to 0.25 m/s (the terminal velocity)	Throw should reach 75% of the way to the nearest wall or obstacle
Drop	Vertical distance the air jet falls below the diffuser before reaching terminal velocity — important in high-ceiling spaces	Drop must not reach occupancy zone (seated: 1.8m above floor) to avoid cold drafts on occupants
Spread	The width of the airflow pattern — how wide the air jet fans out from the diffuser	Spread is controlled by blade angle — wider spread for large open spaces
Neck Velocity	Air speed at the diffuser neck (connection to duct). Directly controls noise generation.	2–4 m/s for offices (NC 30-35). <2 m/s for bedrooms, boardrooms (NC 25)

CAV vs VAV Systems and Air Change Rates

The two fundamental strategies for delivering air to a building differ in how they match supply to the actual cooling or heating demand of the space. Understanding CAV vs VAV determines your entire system design approach.

Figure 3 — CAV vs VAV: The Two Fundamental Air Distribution Control Strategies

CAV — CONSTANT AIR VOLUME

- Fixed airflow rate — always the same CFM
- Temperature varies to control comfort
- Simpler: no VAV boxes or controls
- Energy-inefficient at part load
- Best: small buildings, single zones

Airflow: ■■■■■■■■ (constant)

VAV — VARIABLE AIR VOLUME

- Airflow varies with cooling/heating load
- Temperature stays constant (supply temp)
- VAV boxes modulate airflow per zone
- Energy-efficient — saves 30-50% vs CAV
- Best: large buildings, multi-zone systems

Airflow: ■■■■■■ (varies with load)

Space Type	Recommended ACH (Air Changes/Hour)	Min. Fresh Air (per person)	Notes
General office	6–10 ACH	10 L/s per person	VAV system preferred — load varies significantly between morning and afternoon
Meeting / conference room	8–12 ACH	10 L/s per person	Occupancy-based control (CO■ sensor) saves energy when room is not full
Hospital ward (general)	6–12 ACH	8 L/s per person	All air must be exhausted — no recirculation allowed per HTM 03-01
Operating theatre	20–25 ACH	100% fresh air	Laminar flow at 0.3 m/s min. Positive pressure. HEPA filtration required.
Pharmacy / lab	15–20 ACH	10 L/s per person	Higher ACH for chemical fume control. May need 100% exhaust.
Toilet / bathroom	6–10 ACH (exhaust)	—	Exhaust only — no supply. Timer or occupancy sensor control.
Data centre / server room	25–30 ACH	—	Cooling dominant — often 100% recirculated with computer room AC units (CRAC)
Classroom	6–8 ACH	8 L/s per person	Demand-controlled ventilation (DCV) using CO■ sensors is highly recommended

Common Air Distribution Problem	Cause	Solution
Cold drafts on occupants at desks	Diffuser throw is too long or drop is too great — supply air reaches occupied zone before warming up	Reduce airflow, adjust diffuser blade angle, or use a diffuser with a shorter throw rating for the ceiling height
Hot spots — some areas never cool	Diffuser throw too short — conditioned air does not reach the far zone. Or supply and return too close to each other (short-circuit).	Relocate return grille away from supply, or add a supplementary diffuser to cover the dead zone
Noise complaints from diffusers	Neck velocity too high (> 4 m/s). Duct static pressure too high causing turbulence at diffuser.	Reduce system static pressure. Upsize duct or diffuser. Add attenuator upstream of diffuser.
VAV boxes hunting (cycling open/closed)	VAV box minimum airflow set too low, or zone thermostat too sensitive, or AHU supply temperature too cold.	Raise minimum airflow setpoint. Reset AHU supply air temperature upward at part load.

What comes next — HVAC-009: Psychrometrics — the science of moist air. Dry bulb temperature, wet bulb temperature, relative humidity, specific enthalpy, and dew point. How to read the psychrometric chart and use it to size HVAC coils and understand dehumidification. Essential for all HVAC load calculations.